Lesson 9: Genetics NEA Task 1: Huntington's Disease Inheritance

Pass Criteria P3, P4, P11, P12 & NEA Portfolio Task 1 Preparation

OCR AAQ Level 3 Human Biology (Unit F172) · Synoptic Revision & Clinical Case Study (Ben)

- CAG Triplet Repeat Pathology
- Neuronal Functioning
- Genetic Anticipation
- Autosomal Dominant Pedigree Risk
- Academic Research & Referencing

Molecular Pathology & Anticipation

Clinical Case Study (Ben)

Academic Research & Execution

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Prior Knowledge Retrieval Grid

Genotype of an individual with a heterozygous autosomal dominant disorder?

Hh (Carrying one mutant expanded HTT allele).

Transmission risk when an affected heterozygous parent (Hh) mates with an unaffected partner (hh)?

50% (1 in 2) risk per pregnancy, independent of previous children.

Which trinucleotide sequence expands in exon 1 of the HTT gene on chromosome 4p16.3?

Cytosine-Adenine-Guanine (CAG) (Encoding an elongated polyglutamine tract).

What is the clinical full-penetrance CAG repeat threshold for Huntington's disease?

≥ 40 CAG repeats confers 100% lifetime disease penetrance.

Which specific neuronal population in the basal ganglia undergoes selective apoptosis?

GABAergic Medium Spiny Neurons (MSNs) within the striatum.

Why is genetic anticipation more pronounced during paternal transmission?

Continuous cell divisions during spermatogenesis drive high replication slippage.

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✓ ASSESSMENT REVIEW • MINI-MOCK NEA (GENE EXPRESSION & MRNA VACCINES)

Whole-Class Feedback: What Went Well

Secure understanding of molecular gene expression and foundational clinical-trial methodology

Assessment Focus: Unit F172 NEA
Criteria: P1, P2, P3, P4, M1

🧬 Strong Biological Understanding

DNA → Transcription → mRNA → Ribosome Translation → Amino Acid Sequence → Protein

- **Cytoplasmic Translation:** Most learners accurately explained that vaccine mRNA is translated exclusively in the cytoplasm by ribosomes, requiring *no nuclear entry* and causing *zero alteration to genomic DNA*.
- **Unprotected mRNA Instability:** Strong scientific justifications for why naked mRNA cannot be administered: rapid enzymatic degradation by circulating *RNases*, physiological instability, large molecular size, and negative electrostatic charge.
- **Lipid Nanoparticles (LNPs):** Highly effective explanations linking LNPs to protecting labile mRNA, facilitating cellular endocytosis across the phospholipid membrane, and releasing functional transcripts into cytosol.

📊 Good Use of Clinical-Trial Evidence

- **Rigorous Trial Design Features:** Learners successfully identified gold-standard trial parameters: *large sample sizes* (tens of thousands), *random allocation*, *placebo controls*, and *double-blinding* to minimise bias.
- **Quantitative Efficacy Verification:** Frequent and accurate use of infection rate disparities between vaccinated and placebo cohorts to substantiate strong conclusions regarding short-term vaccine efficacy.
- **Association vs Causation:** High-performing scripts recognised that an adverse medical event occurring sequentially *after* vaccination does not automatically establish physiological *causation*.
- **Balanced Critical Appraisal:** Multiple responses effectively weighed strong primary efficacy evidence against inherent trial limitations (restricted longitudinal follow-up and low statistical power for ultra-rare adverse events).

TEACHER EVALUATION

Assessment Takeaway: Biological comprehension of transcription, translation, and vector delivery is secure across the cohort. The key progression step is translating this descriptive mastery into **evidence-led analysis and critical evaluation (P3, P4, M2, D2)**.

TARGET FOR EXCELLENCE • MOVING TO MERIT & DISTINCTION (P3, P4, M2, D1–D2)

Whole-Class Feedback: Even Better If

Advancing from descriptive narrative to evidence-led analysis, scientific precision, and academic referencing

Key NEA Thresholds:
M2 Analysis · D2 Judgement

Scientific Precision

- **Transcription vs Translation:** Clearly distinguish nuclear DNA transcription from cytoplasmic translation of vaccine mRNA.
- **Base-Pairing Location:** tRNA anticodons align with complementary *codons on mRNA*, never with genomic DNA.
- **Exact Chemical Terms:** Routinely use *transcribed*, *amino-acid sequence*, *peptide bonds*, and *clinical efficacy*.
- **Critical Epistemology:** Never state a trial 'eliminates' chance or bias. Say trial design *minimises bias* and makes chance a *substantially less probable explanation*.

Analysis & Evaluation (M2/D2)

EVIDENCE	What do the primary trial data / findings show?
EXPLANATION	Why does the methodology (e.g. blinding) make this evidence strong?
LIMITATION	What can this clinical design *not* establish or control?
JUDGEMENT	How convincing is the evidence base overall?

> *"The trial provides convincing evidence of short-term efficacy and supports safety due to its randomised, double-blind, placebo-controlled design. However, it cannot demonstrate long-term safety or exclude ultra-rare adverse events."*

Presentation & Referencing

- **Consistent In-Text Citations:** Insert *(Author, Year)* immediately following factual claims, with a complete reference list at the end.
- **Alphabetical Bibliography:** Order full reference entries alphabetically by lead author surname; do *not* number references.
- **Proofreading Rigour:** Audit for sentence length, spelling, and consistent biological terminology throughout.
- **Criteria Headings:** Structure portfolios with explicit headings mapped to assessment criteria for clear assessor verification.

P1–P4 Pass · M2 Merit · D1–D2 Distinction

Clinical Case Study Profile: Ben (40)

Patient Profile

- **Age/Sex:** 40-year-old male.
- **Occupation:** Teacher.
- **History:** Adopted at age 2. Biological family medical history completely unknown. Teenage search unsuccessful.

Diagnostic Journey

- **Primary Care:** GP recognized chorea.
- **Radiology:** Brain MRI confirmed neurodegenerative disease.
- **Confirmation:** Genetic test positive for Huntington's disease.

Clinical Presentation

- **Onset:** Symptoms began 1 year ago (age 39).
- **Motor Control:** Involuntary jerky movements (chorea) in arms, legs, face.
- **Impact:** Impaired hand grip prevents holding a pen; severe embarrassment in the classroom.

Family Context

- **Spouse:** Married (wife is unaffected).
- **Children:** 6-year-old daughter, 3-year-old son.
- **Pregnancy:** Wife is currently pregnant with their 3rd child.
- **Primary Concern:** Devastated about potential genetic transmission to his children.

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The Diagnostic Pathway: From Symptom to Sequence

Step 1: Clinical Observation (GP)

- **Trigger:** Recognition of chorea (abnormal involuntary movements).
- **Action:** Referral to neurology.



Step 2: Structural Imaging (Radiology)

- **Tool:** Brain MRI.
- **Finding:** Bilateral striatal atrophy (shrinkage of caudate nucleus and putamen).
- **Finding:** Lateral ventricle enlargement due to tissue loss.



Step 3: Molecular Confirmation (Clinical Genetics)

- **Tool:** Predictive/Diagnostic Blood Test.
- **Target:** HTT gene on chromosome 4p16.3.
- **Result:** Confirmed CAG repeat count ≥ 40.

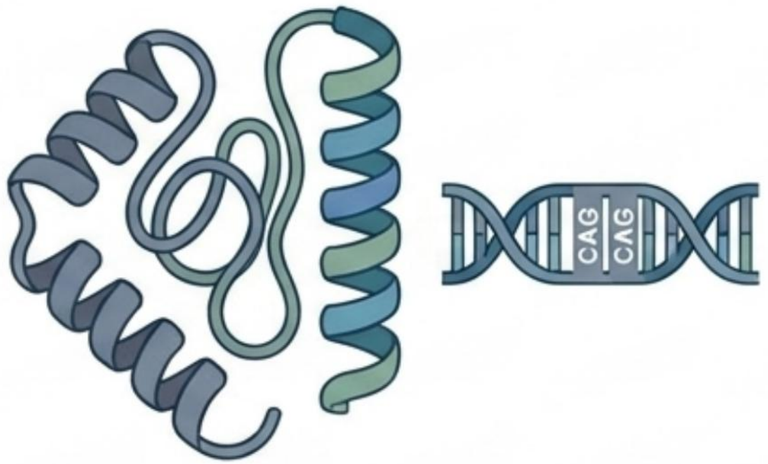


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Molecular Pathology: The HTT Gene & CAG Expansion

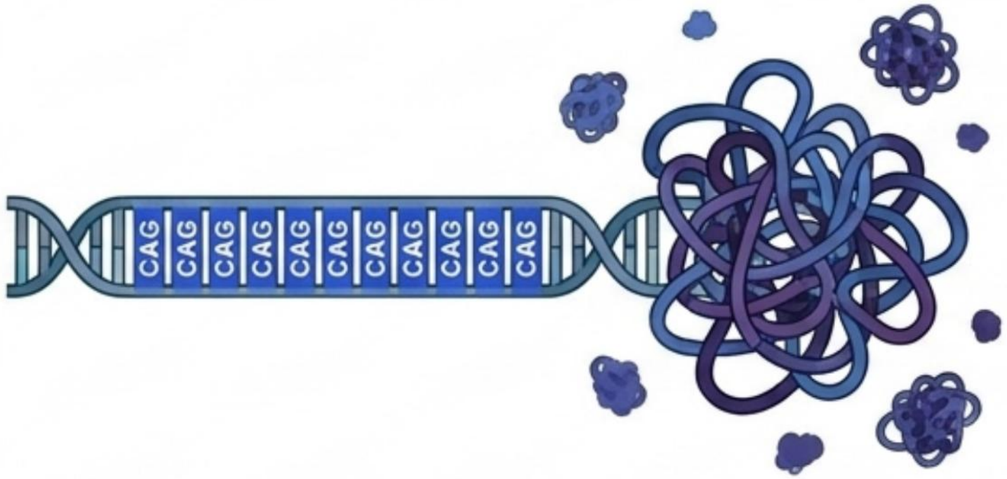
Normal HTT Allele

- **Location:** Chromosome 4p16.3
- **Repeat Count:** < 35 CAG repeats
- **Protein Product:** Standard polyglutamine tract
- **Result:** Healthy Huntingtin protein folding and normal cellular transport.



Mutant HTT Allele

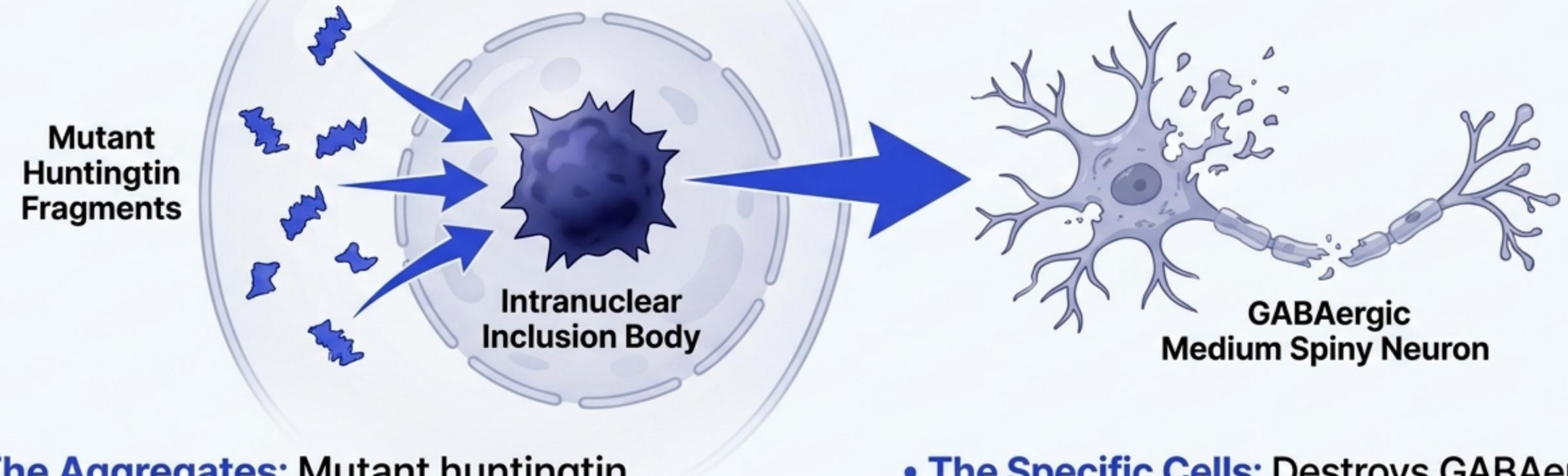
- **Location:** Chromosome 4p16.3
- **Repeat Count:** ≥ 40 CAG repeats
- **Protein Product:** Elongated polyglutamine tract
- **Result:** Protein misfolding, aggregation, and intranuclear inclusions.



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Cellular Degradation: Toxic Gain-of-Function

- **The Mechanism:** Unlike many genetic diseases caused by a loss of function, HD is driven by a **toxic gain-of-function.**
- **The Target:** Causes selective **apoptosis** (programmed cell death) in the basal ganglia.



- **The Aggregates:** Mutant huntingtin fragments accumulate and form toxic intranuclear inclusions.
- **The Specific Cells:** Destroys GABAergic Medium Spiny Neurons (MSNs) in the caudate and putamen.

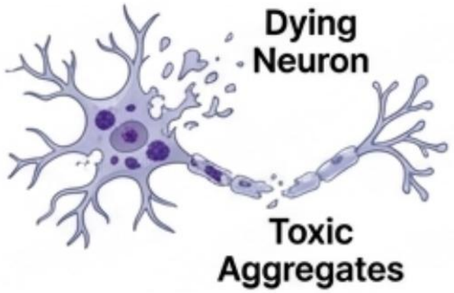
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Synthesis: Mapping the Molecular Defect to Macro Symptoms



The Genetic Error

CAG triplet expansion (≥ 40 repeats) in the HTT gene alters primary protein structure.



The Cellular Death

Toxic polyglutamine aggregates destroy GABAergic MSNs in the striatum.



The Clinical Phenotype

Loss of inhibitory GABA signals removes control over motor pathways.

Result: Ben's chorea (inability to inhibit jerky arm/hand movements).

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Autosomal Dominant Transmission Risk

The Genotype Cross

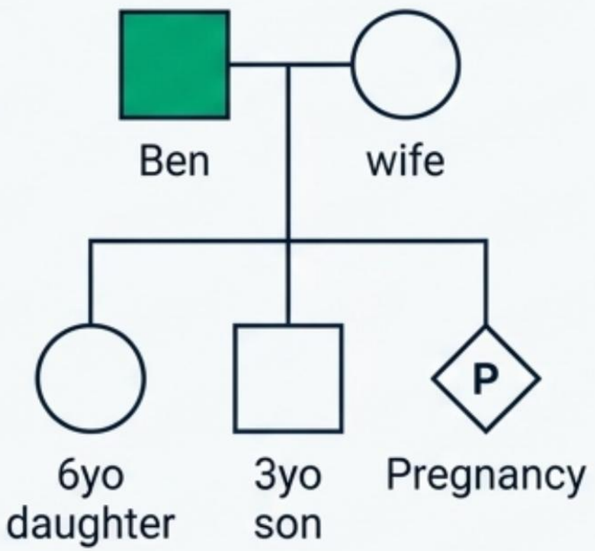
Ben's Genotype:
Heterozygous (Hh)

Wife's Genotype:
Homozygous Recessive (hh)

	H	h
h	Hh	hh
h	Hh	hh

The Punnett Result: 50% Hh, 50% hh.

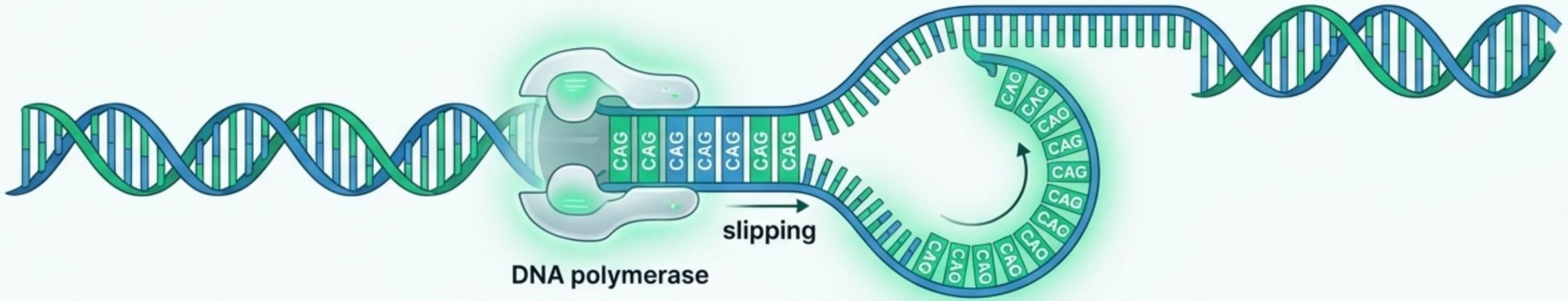
The Family Reality



- **The Rule:** A strict 50% (1 in 2) transmission risk.
- **The Caveat:** Risk is entirely independent for each pregnancy.
- **The Reality:** The 6-year-old, 3-year-old, and unborn child each face this exact 50% probability.

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Genetic Anticipation & Paternal Meiotic Slippage



The Mystery

Why did Ben have no known family history? His biological parent may have died early, been misdiagnosed, or had a later onset.

The Mechanism

DNA polymerase struggles with highly repetitive sequences.

Meiotic Slippage

During spermatogenesis, continuous cell divisions cause the polymerase to "slip" and add extra CAG repeats.

Genetic Anticipation

Paternal transmission often expands the CAG tract, causing earlier symptom onset and increased severity in the next generation.

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Addressing Ben’s Concerns: Testing & The Future

Diagnostic & Predictive Testing (P12)

- **Ben:** Diagnosed via blood test counting CAG repeats.
- **His Children:** Predictive testing for minors is highly restricted due to severe ethical and psychological implications.

Prenatal Testing Options

- **Chorionic Villus Sampling (CVS)** or **Amniocentesis** can test the current pregnancy.
- Raises profound **ethical questions** regarding continuation of pregnancy based on incurable genetic results (P10).

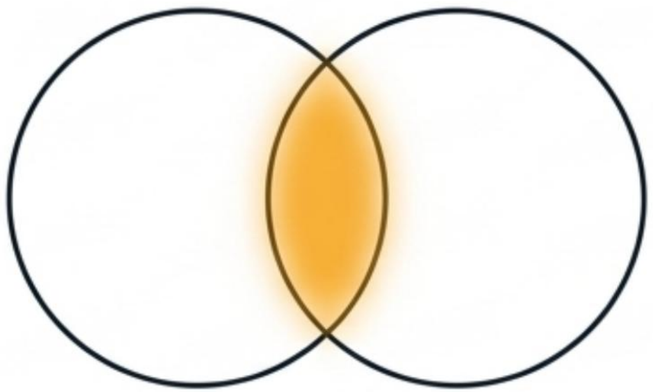
Emerging Medical Technologies

- **Currently**, GP manages symptoms only; no cure exists.
- **Future hope:** Gene therapy and CRISPR-Cas9 trials aiming to 'silence' the mutant HTT allele.

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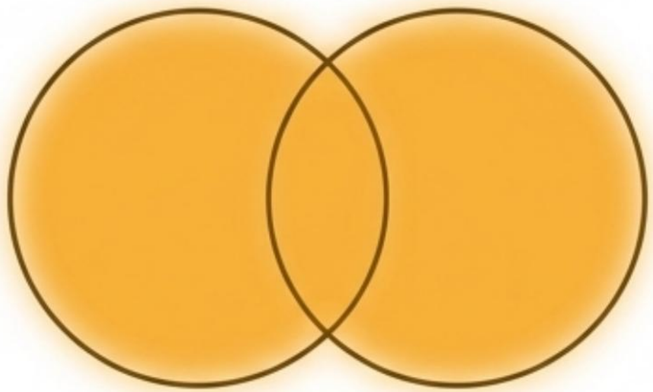
Academic Skills: Boolean Literature Searching

AND



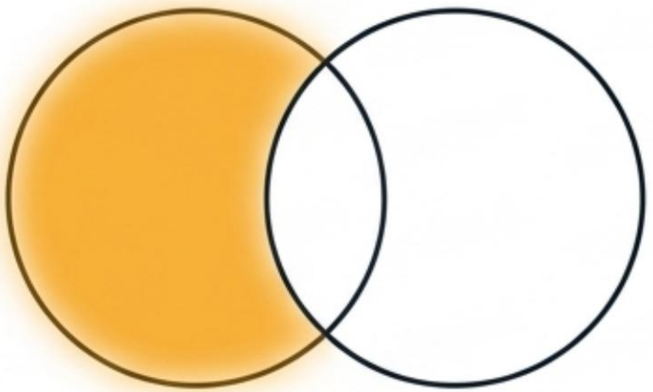
Function: Narrows results. Both terms must be present.

OR



Function: Broadens results. Connects synonyms (e.g., Chorea OR Dyskinesia).

NOT



Function: Excludes specific terms to filter noise.

Actionable Example (Search String):

```
("Huntington's Disease" OR "HD") AND ("CAG repeat" OR "Medium Spiny Neuron")
```

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Source Evaluation: The CRAAP Framework

C	**Currency:**	Is the research recent enough for rapid-moving genetic science? (Post-2015).
R	**Relevance:**	Does it specifically address molecular pathology or inheritance?
A	**Authority:**	Is the author a qualified geneticist, neurologist, or major institution?
A	**Accuracy:**	Is the data peer-reviewed and supported by clinical evidence?
P	**Purpose:**	Is the source objective, or is it biased/selling a product?

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Academic Integrity: Deconstructing Harvard Referencing



Zero Tolerance: Plagiarism (including undisclosed AI usage) invalidates your NEA. Always cite your sources.

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NEA Task 1 Blueprint: The Top-Marks Checklist

Biological Mechanisms (The 'What')

- [ ] **P3:** Detail the genetic variation (CAG triplet expansion ≥ 40).
- [ ] **P3/P9:** Connect toxic gain-of-function (MSN death) to symptoms (Chorea/Mental impact).
- [ ] **P4/P11:** Accurately diagram autosomal dominant inheritance (50% risk).
- [ ] **P12:** Explain diagnostic genetic testing methods.

Academic Execution (The 'How')

- [ ] Utilize Boolean search strings to find high-level journals.
- [ ] Apply the CRAAP framework to evaluate source credibility.
- [ ] Use flawless Harvard referencing for all data and claims.
- [ ] Ensure 100% original synthesis to maintain academic integrity.

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